

Jiaqi Yao

☎ 1 773-961-4630 ✉ trevoryao4@gmail.com in LinkedIn/jiaqi yao 🏠 Evanston, Illinois, USA

Education

Northwestern University Doctor of Philosophy in Biomedical Engineering GPA: 3.9/4.0	Evanston, USA Sep 2025 – present
The University of Adelaide (UOA) Bachelor of Science in Mathematical Sciences (Honours) GPA: 7.0/7.0 First Class Honours	Adelaide, Australia Jul 2023 – Jun 2025
Ocean University of China (OUC) Bachelor of Science in Mathematics and Applied Mathematics GPA: 3.9/4.0 WAM: 93.5/100	Qingdao, China Sep 2021-Jul 2023

Publication

- [1] **Yao, J.**, Mitchell, L, Maclean, J, Saratchandran, H. (2025). Data Denoising and Derivative Estimation for Data-Driven Modeling of Nonlinear Dynamical Systems, arXiv:2509.14219. <https://arxiv.org/abs/2509.14219>
- [2] **Yao, J.**, Liu, B. (2024). Community-Detection Method of Complex Network Based on Node Influence Analysis. *Symmetry*, 16(6), 754. <https://doi.org/10.3390/sym16060754>
- [3] **Yao, J.**, Mitchell, L. (2024). Community Detection of Complex Network Based on Graph Convolution Iterative Algorithm, arXiv:2412.02711. <https://arxiv.org/abs/2412.02711>
- [4] Jiang, Z., **Yao, J.**, Meng, L., Wang, J., Cao, Y. (2023). Research of the Propagation Mechanism of Scream Effect and its Simulation. *Mathematical Modelling & Its Applications*, 12, 64-72+106. <https://doi.org/10.19943/j.2095-3070.jmmia.2023.04.09>

Awards

Major Awards

2025 University Medal The University of Adelaide <i>Granted to the top 18 graduating students for highest academic distinction</i>	Jan 2026
Finalist Award of Mathematical Contest in Modeling/Interdisciplinary Contest in Modeling (MCM/ICM) Consortium for Mathematics and Its Applications <i>Placed among the top 1% of teams worldwide in an international mathematical modeling competition</i>	May 2024
National Second Prize of Contemporary Undergraduate Mathematical Contest in Modelling (CUMCM) Chinese Society for Industrial and Applied Mathematics <i>Awarded at the national level for excellence in mathematical modeling</i>	Nov 2022
China National Scholarship for Undergraduate Students Ministry of Education of the People's Republic of China <i>Highest national scholarship to academically outstanding undergraduates</i>	Dec 2022

Selected Awards

The Executive Dean's Recognition of Academic Excellence The University of Adelaide	Apr 2025
David Murray Prize in Mathematical Sciences The University of Adelaide	Apr 2025
RB Potts Prize The University of Adelaide	Apr 2025

School of Computer and Mathematical Sciences Charles EM Pearce Prize The University of Adelaide	Apr 2025
AIML Research Scholarship Australian Institute for Machine Learning (AIML)	Jul 2024
Summer Research Scholarship The University of Adelaide	Jan 2024
Bank of Qingdao Outstanding Student Scholarship Bank of Qingdao	Nov 2023
First Prize Scholarship Ocean University of China	Sep 2023
Global Citizen Scholarship The University of Adelaide	Jul 2023
National Third Prize of “Shenzhen Cup” Mathematical Modeling Challenge Chinese Society of Industrial and Applied Mathematics	Dec 2022
First Prize Scholarship Ocean University of China	Sep 2022
First-class Outstanding Student Ocean University of China	Sep 2022
Third Prize of Mathematical Modelling Competition of Ocean University of China Ocean University of China	Jun 2022
National Second Prize of “Mathorcup” University Mathematical Modeling Challenge Chinese Society of Optimization, Overall Planning and Economic Mathematics	May 2022

Work Experience

Research Assistant Northwestern University, XIQI Lab <i>Advisor: Prof. Xi Chen</i>	Jul 2025-Aug 2025
- Developed a novel phase-sensitive optical imaging simulation based on the finite-difference time-domain (FDTD) method. - Discovered theoretical physics models to explain coherence-based imaging techniques. - Designed machine learning models that enable label-free, in vivo three-photon imaging.	
Research Assistance The University of Adelaide, Australian Institute for Machine Learning <i>Advisors: Prof. Lewis Mitchell, Dr. Hemanth Saratchandran, Dr. John Maclean</i>	Jul 2024-Jun 2025
- Proposed a novel sampling strategy for physics-informed neural networks (PINNs) to improve model accuracy. - Designed the RKTv-INR framework, an efficient method for denoising dynamical system data and estimating temporal derivatives, enabling more accurate system identification. - Developed an adaptive function library for sparse identification of nonlinear dynamics (SINDy) based on implicit neural representations.	
Associate Lecturer The University of Adelaide	Apr 2024
- Contributed to the instruction of undergraduate courses Science Processes and Practices, Numerical Methods, and Optimisation and Operations Research by leading tutorials, designing problem sets, and mentoring students. - Coordinated academic collaborations between Chinese and Australian faculty members. - Participated in the design of course ‘Science Processes and Practices’ assessments.	
Summer Research Project The University of Adelaide	Nov 2023 – Jan 2024

- Introduced a data sciencebased framework to evaluate the equity of urban green space distribution
- Designed and implemented a parallel computing pipeline to analyze Adelaides regional division, green space distribution, and transportation network data, revealing spatial inequalities in green space accessibility.
- Delivered a research presentation on Adelaides green space distribution at the Summer Research Conference hosted by the Australian Mathematics Science Institute (AMSI).

Student Research Developing Program

Nov 2022 – Nov 2023

Ocean University of China

- Designed a fast k -core decomposition algorithm for a specific class of graphs, such as ER graphs and hypergraphs, based on their unique characteristics.
- Constructed a large-scale social network model and built the SEIR model.
- Proposed a new mathematical model that simulate the outbreak and propagation of information.

Key Skills

Programming:

Python, MATLAB, Julia, C++, R

Language:

English (proficient), Chinese (Mandarin, native)

Mathematical Analysis:

Fourier Optics, Signal Processing, Machine Learning, Differential Equations, Numerical Simulation, Optimization

Internship

China Coal Trading Data Network (CCTD)

Beijing, China

Data Science Internship

Jan 2023 – Feb 2023

- Built and automated data-processing pipelines to aggregate and visualize large-scale historical anthracite market data, including production, consumption, imports/exports, inventories, and transportation flows.
- Developed deep learning models to forecast anthracite price trends using multi-source market indicators.
- Enhanced the CCTD web platform by optimizing data presentation and analytical content for improved user engagement.

Extracurricular Activities

Industry Consultation on Microscopy System Design

Dec 2025

- Participated in an online technical consultation with medical device companies, presenting fluorescence microscopy and quantitative phase imaging (QPI) technologies to support imaging module design.
- Provided system architecture guidance and equipment procurement recommendations.

Student Ambassador at 2024 UoA China Open Day

Apr 2024

- Represented the University of Adelaide at the 2024 China Open Day, sharing academic experiences and advising prospective students on study pathways and program opportunities.

Interview for Strategic Partnership between UOA and Haide College of OUC

Oct 2023

- Contributed to discussions on institutional collaboration, providing analytical feedback and strategic recommendations for the development of Haide College.

Youth Volunteer Association Leader

Sep 2022-Jul 2023

- Led and coordinated community outreach initiatives, organizing large-scale volunteer activities including public anti-drug awareness campaigns.